

$$\begin{Bmatrix} \Delta\beta \\ \Delta\bar{p} \\ \Delta\bar{r} \\ \Delta\xi_y \\ \Delta\phi \\ \Delta\psi \end{Bmatrix} = C \begin{Bmatrix} 0.000330 \\ 0.000000 \\ 0.000236 \\ 0.999880 \\ 0.000000 \\ -0.015512 \end{Bmatrix} \exp(-0.015184 \tau_y)$$

The details of this numerical comparison are left as an exercise for the reader.

Clearly, the spiral mode approximation expressed in Eq. (9.4.11) represents a tremendous improvement over Eq. (9.4.5). The most serious inaccuracy in the approximations associated with obtaining Eq. (9.4.5) is neglecting the effects of bank angle. The bank angle component of the eigenvector for spiral motion is one to two orders of magnitude larger than either the sideslip angle or the dimensionless yawing rate, both of which were retained in the earlier approximation. In fact, the change in bank angle provides the primary driving force associated with spiral motion.

Comparing the spiral mode damping rate from Eq. (9.4.6) with that from Eq. (9.4.12) shows some similarity in dependence on the stability and damping derivatives. However, these two approximations show a very different dependence on other aircraft design and operating parameters. The spiral approximation given by Eq. (9.4.6) predicts a damping rate that is proportional to airspeed, whereas Eq. (9.4.12) shows an inverse relation between airspeed and spiral mode damping rate. Equation (9.4.6) predicts a spiral mode damping rate that is directly proportional to air density, wing area, and the square of the wing span, while being inversely proportional to the yawing moment of inertia. On the other hand, Eq. (9.4.12) predicts no dependence on any of these parameters. Results obtained from Eq. (9.4.12) are much more consistent with reality.

EXAMPLE 9.4.1. The general aviation airplane described in Example 9.2.1 is modified so that the directional stability is doubled,

$$C_{n,\beta} = +0.140$$

All other parameters affecting lateral motion are unchanged. For this modified airplane, use Eq. (9.2.2) to find the eigenvalue and eigenvector corresponding to the spiral mode. Compare this exact solution with the spiral mode approximation given by Eq. (9.4.11).

Solution. For the modified airplane, we have

$$\begin{aligned} R_{\ell,\beta} &\equiv \frac{\rho S_w b_w^3}{8I_{xxb}} C_{\ell,\beta} = -0.1481, & R_{n,\beta} &\equiv \frac{\rho S_w b_w^3}{8I_{zzb}} C_{n,\beta} = 0.0790, \\ R_{\ell,\bar{p}} &\equiv \frac{\rho S_w b_w^3}{8I_{xxb}} C_{\ell,\bar{p}} = -0.8099, & R_{n,\bar{p}} &\equiv \frac{\rho S_w b_w^3}{8I_{zzb}} C_{n,\bar{p}} = -0.0325, \end{aligned}$$